

MariaDB [category]> select * from employee;

```
+-----+-----+-----+-----+-----+-----+-----+
-----+
| empNo | fName | lName | address | DOB      | sex | position      |
deptNo |
+-----+-----+-----+-----+-----+-----+-----+
-----+
| E1    | Ama  | Mensah | Accra    | 1985-03-15 | F   | Manager      |
10 |
| E10   | Fiifi | Darko  | Sunyani  | 1990-02-20 | M   | Software Developer |
50 |
| E2    | Kwame | Asante | Kumasi   | 1990-07-22 | M   | Team Leader   |
10 |
| E3    | Akosua | Boateng | Tamale   | 1992-11-10 | F   | Manager      |
20 |
| E4    | Kojo  | Owusu  | Cape Coast | 1995-05-18 | M   | Software Developer |
20 |
| E5    | Efua  | Addo   | Takoradi | 1988-09-25 | F   | Analyst       |
30 |
| E6    | Yaw   | Bediako | Accra    | 1991-01-12 | M   | Analyst       |
30 |
| E7    | Abena | Osei   | Koforidua | 1993-06-30 | F   | Software Developer |
40 |
| E8    | Kofi  | Appiah | Tema     | 1989-09-05 | M   | Team Leader   |
40 |
| E9    | Adjoa | Nyarko | Ho       | 1994-12-11 | F   | Analyst       |
50 |
+-----+-----+-----+-----+-----+-----+-----+
-----+
10 rows in set (0.001 sec)
```

MariaDB [category]> select * from project;

```
+-----+-----+-----+
| projNo | projName      | deptNo |
+-----+-----+-----+
| AI      | AI Assistant  | 40     |
| CRM     | Customer Relationship Management | 40     |
| DBMS    | Database Management System | 20     |
| HR      | Human Resource System | 30     |
| LIB     | Library Management System | 20     |
| NET     | Network Monitoring | 50     |
| PAY     | Payroll System | 10     |
| SCCS    | Student Course Control System | 10     |
| SEC     | Security Monitoring System | 30     |
| WEB     | University Website | 10     |
+-----+-----+-----+
10 rows in set (0.001 sec)
```

```
MariaDB [category]> select * from department;
```

deptNo	deptName	mgrEmpNo
10	Software Development	E1
20	Data Analytics	E3
30	Cyber Security	E5
40	AI Research	E7
50	Networking	E9

```
5 rows in set (0.001 sec)
```

```
MariaDB [category]> select * from workson;
```

empNo	projNo	dateWorked	hoursWorked
E1	SCCS	2026-06-01	8
E1	SCCS	2026-06-02	6
E2	PAY	2026-06-01	7
E3	DBMS	2026-06-01	5
E4	WEB	2026-06-02	8
E5	SEC	2026-06-03	6
E6	HR	2026-06-03	7
E7	AI	2026-06-04	8
E8	CRM	2026-06-04	5
E9	NET	2026-06-05	6

```
10 rows in set (0.001 sec)
```

7.25)

```
MariaDB [category]> create view FemaleManagedProjects as
```

```
-> select
->     p.projNo,
->     p.projName,
->     p.deptNo
-> from Project p
-> join Department d on p.deptNo = d.deptNo
-> join Employee e on d.mgrEmpNo = e.empNo
-> where e.sex = 'F'
-> order by p.projNo;
```

```
Query OK, 0 rows affected (0.033 sec)
```

```
MariaDB [category]> select * from FemaleManagedProjects;
```

projNo	projName	deptNo
--------	----------	--------

AI	AI Assistant	40
CRM	Customer Relationship Management	40
DBMS	Database Management System	20
HR	Human Resource System	30
LIB	Library Management System	20
NET	Network Monitoring	50
PAY	Payroll System	10
SCCS	Student Course Control System	10
SEC	Security Monitoring System	30
WEB	University Website	10

10 rows in set (0.032 sec)

7.26)

```
MariaDB [category]> create view EmployeeProjectView as
-> select
->     e.empNo,
->     e.fName,
->     e.lName,
->     p.projName,
->     w.hoursWorked
-> from Employee e
-> join WorksOn w on e.empNo = w.empNo
-> join Project p on w.projNo = p.projNo;
Query OK, 0 rows affected (0.034 sec)
```

```
MariaDB [category]> select * from EmployeeProjectView;
```

empNo	fName	lName	projName	hoursWorked
E1	Ama	Mensah	Student Course Control System	8
E1	Ama	Mensah	Student Course Control System	6
E2	Kwame	Asante	Payroll System	7
E3	Akosua	Boateng	Database Management System	5
E4	Kojo	Owusu	University Website	8
E5	Efua	Addo	Security Monitoring System	6
E6	Yaw	Bediako	Human Resource System	7
E7	Abena	Osei	AI Assistant	8
E8	Kofi	Appiah	Customer Relationship Management	5
E9	Adjoa	Nyarko	Network Monitoring	6

10 rows in set (0.005 sec)

7.27)

```
MariaDB [category]> create view EmpProject(empNo, projNo, totalHours) as
-> select
->     w.empNo,
->     w.projNo,
->     sum(hoursWorked)
-> from Employee e, Project p, WorksOn w
-> where e.empNo = w.empNo
-> and p.projNo = w.projNo
-> group by w.empNo, w.projNo;
Query OK, 0 rows affected (0.031 sec)
```

7.27 a)

```
MariaDB [category]> select * from EmpProject;
```

```
+-----+-----+-----+
| empNo | projNo | totalHours |
+-----+-----+-----+
| E1    | SCCS   |          14 |
| E2    | PAY    |           7 |
| E3    | DBMS   |           5 |
| E4    | WEB    |           8 |
| E5    | SEC    |           6 |
| E6    | HR     |           7 |
| E7    | AI     |           8 |
| E8    | CRM    |           5 |
| E9    | NET    |           6 |
+-----+-----+-----+
9 rows in set (0.031 sec)
```

7.27 b)

```
MariaDB [category]> SELECT projNo
-> FROM EmpProject
-> WHERE projNo = 'SCCS';
```

```
+-----+
| projNo |
+-----+
| SCCS   |
+-----+
1 row in set (0.005 sec)
```

7.27 c)

```
MariaDB [category]> SELECT COUNT(projNo)
-> FROM EmpProject
-> WHERE empNo = 'E1';
```

```
+-----+
| COUNT(projNo) |
+-----+
|              1 |
```

```
+-----+
1 row in set (0.003 sec)
```

7.27 d)

```
MariaDB [category]> SELECT empNo, totalHours
-> FROM EmpProject
-> GROUP BY empNo;
```

```
+-----+
| empNo | totalHours |
+-----+
| E1    |          14 |
| E2    |           7 |
| E3    |           5 |
| E4    |           8 |
| E5    |           6 |
| E6    |           7 |
| E7    |           8 |
| E8    |           5 |
| E9    |           6 |
+-----+
```

```
9 rows in set (0.002 sec)
```

7.28) A materialized view stores the result of the query physically. It can be maintained when rows are inserted or updated such that a part newly satisfies the condition `partCost > 1000`. But, because the view uses `distinct`, deletions, and some updates may require access to the base table 'Part' to determine whether the part should remain in the materialized view. Thus, maintenance without accessing 'Part' is possible only when the effect of the change can be determined solely from the modified row alone.